

Data Engineering

By Abdul Rehman Zafar

Agenda

- Introduction
- Survey
- Getting to know each other
- Attendance Policy
- Grading Policy
- Course Outline
- Quizzes, labs and projects
- Introduction to data engineering

Introduction

- Bachelors in Software Engineering (GIKI, Pakistan)
- Masters in Distributed Systems (Technische Universität Darmstadt, Germany)
- 15 years in IT industry
- Previous roles
 - Software Engineer
 - Chief Technology Officer
 - Cloud and Data Consulting in CGI
 - **Senior Solutions Architect** in Amazon Web Services (**AWS**) for 5 years
- Current Role
 - **Senior Streaming Solutions Architect** in **Veverica** (Alibaba Cloud) for last 2 years
 - **Original Creators of Apache Flink**
 - Regular Speaker at Big Data conferences
 - Taught Cloud Computing in UE
- Feel free to add me on LinkedIn
 - <https://www.linkedin.com/in/abdulrehmanzafar/>

Survey

Please check Teams Channel and fill in the survey

Getting to know each other

- Name
- Where are you from?
- How many years of work experience do you have in IT?
- What is your expertise based on your work experience?
- Which programming languages do you use?
- Which data engineering tools/frameworks have you used?
- Which topics you would like to learn in this course?
- What is the most interesting thing you have built using AI?
- Any fun fact about you?

Attendance Policy

- On campus classes attendance is mandatory
 - Unless exceptional circumstances, please send the justification via proof in the **email**.

Grading Policy

Quizzes	25%	During Lectures
Labs	45%	Lectures / Homework
Solo Project	25%	Mini 4 hour project
Report	10%	Project Report
Total	100%	

- No written/oral exam
- Minimum passing Grade **50%**
- Late submissions after deadline will be marked **Zero**

Initial Course Outline (To Adapt) - Labs (45%) and Quizzes/Knowledge Checks (25%)

Module 1: Welcome to AWS Academy Data Engineering

- Course prerequisites and objectives
- Course overview

Module 2: Data-Driven Organizations

- Data-driven decisions
- The data pipeline – infrastructure for data-driven decisions
- The role of the data engineer in data-driven organizations
- Modern data strategies
- Lab: Accessing and Analyzing Data by Using Amazon S3
- Knowledge check

Module 3: The Elements of Data

- The five Vs of data – volume, velocity, variety, veracity, and value
- Volume and velocity
- Variety – data types
- Variety – data sources
- Veracity and value
- Activities to improve veracity and value
- Activity: Planning Your Pipeline
- Knowledge check

Module 4: Design Principles and Patterns for Data Pipelines

- AWS Well-Architected Framework and Lenses
- Activity: Using the Well-Architected Framework
- The evolution of data architectures
- Modern data architecture on AWS
- Modern data architecture pipeline: Ingestion and storage
- Modern data architecture pipeline: Processing and consumption
- Streaming analytics pipeline
- Lab: Querying Data by Using Athena
- Knowledge check

Module 5: Securing and Scaling the Data Pipeline

- Cloud security review
- Security of analytics workloads
- ML security
- Scaling: An overview
- Creating a scalable infrastructure
- Creating scalable components
- Knowledge check

Module 6: Ingesting and Preparing Data

- ETL and ELT comparison
- Data wrangling introduction
- Data discovery
- Data structuring
- Data cleaning
- Data enriching
- Data validating
- Data publishing
- Knowledge check

Module 7: Ingesting by Batch or by Stream

- Comparing batch and stream ingestion
- Apache Kafka and Apache Flink
- Batch ingestion processing
- Purpose-built ingestion tools
- AWS Glue for batch ingestion processing
- Scaling considerations for batch processing
- Lab: Performing ETL on a Dataset by Using AWS Glue
- Kinesis for stream processing
- Scaling considerations for stream processing
- Ingesting IoT data by stream
- Knowledge check

Module 8: Storing and Organizing Data

- Storage in the modern data architecture
- Data lake storage
- Data warehouse storage
- Purpose-built databases
- Storage in support of the pipeline
- Securing storage
- Lab: Storing and Analyzing Data by Using Amazon Redshift
- Knowledge check

Module 9: Processing Big Data

- Big data processing concepts
- Apache Hadoop
- Apache Spark
- Amazon EMR
- Managing your Amazon EMR clusters
- Lab: Processing Logs by Using Amazon EMR
- Apache Hudi
- Lab: Updating Dynamic Data in Place
- Knowledge check

Module 10: Processing Data for ML

- ML concepts
- The ML lifecycle
- Framing the ML problem to meet the business goal
- Collecting data
- Applying labels to training data with known targets
- Activity: Labeling with SageMaker Ground Truth
- Preprocessing data
- Feature engineering
- Developing a model
- Deploying a model
- ML infrastructure on AWS
- SageMaker
- Demo: Preparing Data and Training a Model with SageMaker
- Demo: Preparing Data and Training a Model with SageMaker Canvas
- AI/ML services on AWS
- Knowledge check

Module 11: Analyzing and Visualizing Data

- Considering factors that influence tool selection
- Comparing AWS tools and services
- Demo: Analyzing and Visualizing Data with AWS IoT Analytics and QuickSight
- Selecting tools for a gaming analytics use case
- Lab: Analyzing and Visualizing Streaming Data with Kinesis Data Firehose, OpenSearch Service, and OpenSearch Dashboards
- Knowledge check

Module 12: Automating the Pipeline

- Automating infrastructure deployment
- CI/CD
- Automating with Step Functions
- Lab: Building and Orchestrating ETL Pipelines by Using Athena and Step Functions
- Knowledge check

Solo Project / Capstone Project (25% + 10%)

- Small mini project
- Should only take 2 - 4 hours
- To be done by each student individually (No groups)
- No presentation/demo/report needed
- Only Hands on project (25%) and Report submission (10%)
- The Capstone Project provides
 - Integrative project-based learning experience
 - Reinforces technical skills that are taught in this course.
 - Offers students an opportunity to demonstrate
 - Critical thinking,
 - Problem solving,
 - Software development lifecycle
 - Communication skills.

Questions?